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Saveetha Institute of Medical And Technical Sciences

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ITA0329 – MOBILE COMPUTING

Mobile App Performance Optimization Toolkit (Multithreading & UI Rendering).

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ABSTRACT

- Mobile applications may experience lag, UI freezing and slow screen rendering during heavy processing.
- This project proposes a toolkit to monitor performance and improve responsiveness through multithreading and efficient UI rendering.
- Background tasks are separated from the main UI thread to reduce blocking and improve task execution.
- Performance is evaluated by comparing execution time, CPU and memory usage, frame rate and UI responsiveness before and after optimization.

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INTRODUCTION

- A mobile application performance toolkit that observes app behaviour and demonstrates optimization techniques for background processing and UI rendering.
- Responsive applications improve usability. Efficient thread management prevents long-running work from blocking the interface.
- Reduces UI freezing and rendering delays, providing a smoother and better user experience.
- Android applications • Mobile games • E-commerce apps
- Real-time dashboards • Apps that process images, data or network requests

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PROBLEM STATEMENT

- The project addresses poor performance monitoring, inefficient main-thread task execution, and UI rendering issues that cause application bottlenecks, screen freezing, dropped frames, and reduced responsiveness.

OBJECTIVES

- To monitor and analyze mobile application performance using execution-time and resource measurements.
- To implement multithreading and background task execution to reduce main-thread workload.
- To optimize UI rendering and evaluate improvements in frame rate and application responsiveness.

EXISTING SYSTEM

- Time-consuming operations may run on the main thread.
- Heavy processing can cause UI delays or temporary screen freezing.
- Performance issues may be difficult to identify without centralized monitoring.
- Inefficient rendering can result in frame drops and slower screen transitions.

PROPOSED SYSTEMS

- Uses multithreading to move suitable time-consuming tasks away from the main UI thread.
- Monitors execution time, CPU usage, memory consumption and rendering behaviour.
- Applies UI rendering improvements and compares performance before and after optimization.

Modules

- **Module 1 – Performance Monitoring and Analysis**: Monitors CPU usage, memory consumption, frame rate and execution time to identify performance bottlenecks.

  Outcome: Detects resource-related issues and provides performance insights.
- **Module 2 – Multithreading and Task Optimization**: Uses Kotlin Coroutines or Executor Service for background task execution and parallel processing.

  Outcome: Reduces UI blocking and improves task execution efficiency.
- **Module 3 – UI Rendering and Performance Optimization**: Optimizes RecyclerView, image caching and lazy loading to reduce frame drops.

  Outcome: Provides smoother UI rendering and better application responsiveness.

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